Project Title

SOA Project Review-1

Date& Time

Under the Guidance of

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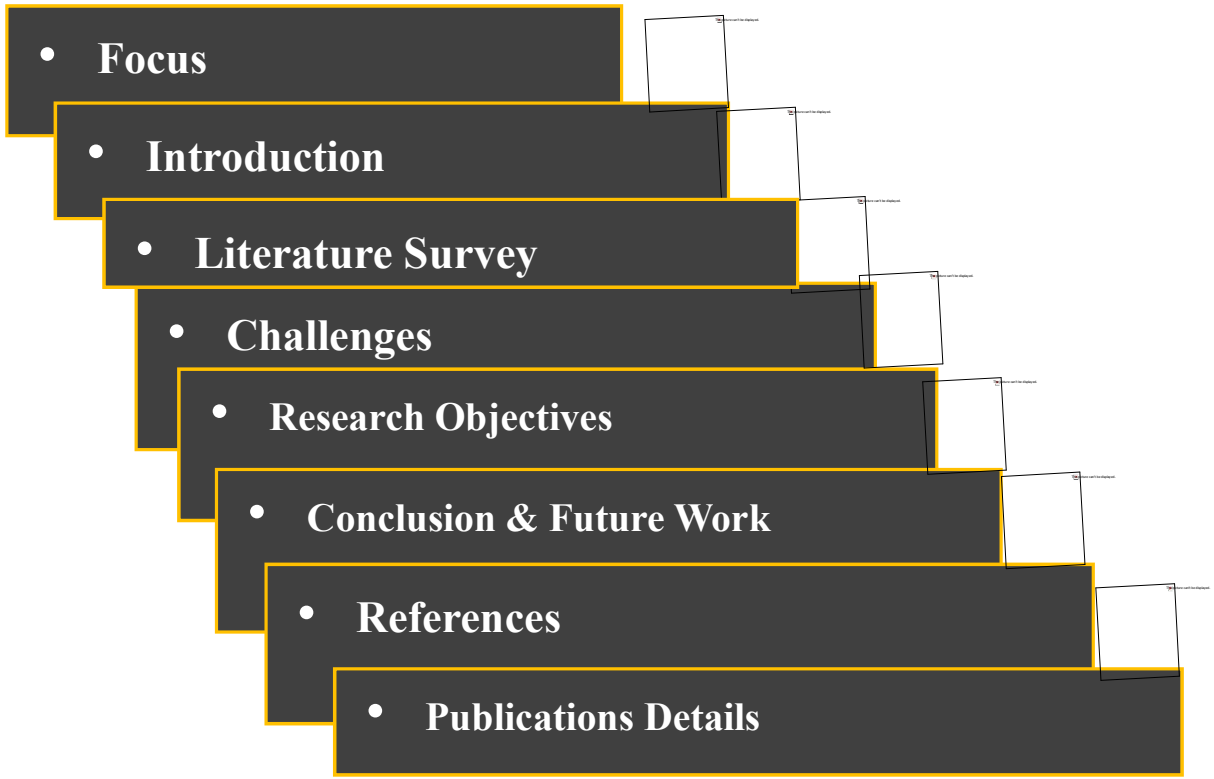
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Outline of Project Work



Problem Statement

AI-Powered Skill Matching Recruitment System

- Manual screening of large numbers of resumes is **time-consuming.**
- Traditional recruitment processes are **difficult to scale.**
- Manual evaluation can lead to **inconsistent candidate assessment.**
- Keyword-based systems may fail to recognize **equivalent skills expressed using different terms.**
- Relevant candidate experience may be missed when it is described in **different contexts.**
- There is a need for an **intelligent recruitment system** to automatically analyse resumes and job requirements.
- The system should **identify relevant skills** and calculate **semantic similarity** between candidates and jobs.
- The system should **rank suitable candidates** based on their skill match.
- The system should provide **explainable matching results** to help recruiters make better decisions.

Objectives

- To develop an **AI-powered system for automated resume and job-description analysis.**
- To extract relevant **skills, education, certifications, and experience** using NLP techniques.
- To perform **semantic skill matching** between candidates and job requirements instead of relying only on keywords.
- To generate an **explainable compatibility score** showing matched and missing skills.
- To automatically **rank and recommend suitable candidates/jobs.**
- To implement the system using a **scalable microservices architecture.**
- To reduce the time and effort required for the initial recruitment screening process.

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Roy et al. [1]	A Machine Learning approach for automation of Resume Recommendation System	• Resume classification followed by content-based recommendation using **cosine similarity and k-NN.**	• Automated classification and matching can reduce manual resume screening and rank candidates according to job-description similarity.	Mainly focuses on traditional ML and similarity-based matching. Future systems can improve semantic understanding using transformer-based NLP models.
Huang et al. [2]	Talent Recommendation Based on Attentive Deep Neural Network and Implicit Relationships of Resumes	• Uses an **attentive deep neural network** to model relationships between resumes and recruitment requirements.	• Deep learning can capture relationships within recruitment data and improve talent recommendation beyond simple keyword matching.	Dynamic changes in the importance of different candidate/job features remain challenging.
Vyas et al. [3]	JobMatchAI: An Intelligent Job Matching Platform Using Knowledge Graphs, Semantic Search and Explainable AI	• Combines **Transformer embeddings, skill knowledge graphs, semantic search, and interpretable reranking.**	• Semantic matching can handle skill synonyms and nonlinear career paths while providing factor-wise explanations.	• Future work can further improve scalability, real-world evaluation, and integration with recruitment workflows.

Problem Gap Identification

- Existing recruitment systems often rely heavily on **keyword-based matching.**
- Keyword matching may fail to recognize **skill synonyms and contextual relationships.**
- Many systems provide a **matching score without clearly explaining why** a candidate received that score.
- Resume information is usually **unstructured,** making automated extraction challenging.
- Existing approaches may focus on either **resume screening or job recommendation,** rather than providing an integrated recruitment workflow.
- There is a need for a **scalable microservices-based architecture** that separates authentication, resume processing, job management, matching, and recommendation services.
- A major gap is the combination of **semantic skill matching + explainability + skill-gap identification + scalable architecture** in one system.

One-line gap statement

Innovation, Creativity, and Novelty

1. Semantic Skill Matching

Instead of simply checking whether two texts contain the same keywords, the system understands the semantic relationship between skills and job requirements using transformer-based embeddings.

2. Skill Synonym Recognition

The system can recognize relationships such as:

"Machine Learning" ↔ "ML"

"JavaScript" ↔ "JS"

"React.js" ↔ "React"

This helps avoid rejecting candidates simply because different terminology is used.

3. Explainable Matching

Instead of displaying only:

Match Score: 82%

the system can show:

Matched Skills: Java, Spring Boot, SQL

Missing Skills: Docker, AWS

Experience Match: High

Overall Match: 82%

4. Skill-Gap Identification

The system identifies the skills a candidate lacks for a particular job and can suggest

Feasibility and Project Plan

Technical Feasibility

- React.js + Tailwind CSS — Frontend
- Java + Spring Boot — Backend
- Python + NLP/Transformer Models — AI Matching
- MySQL/PostgreSQL — Database
- REST APIs + JWT — Security & Communication

Operational Feasibility

- Candidate resume upload and profile management
- Recruiter job posting and applicant management
- AI-based candidate ranking and skill matching
- Matched and missing skills displayed for explanation

Economic Feasibility

- Uses open-source technologies
- Low development and maintenance cost
- Cloud deployment can be added when required

Timeline

W1 → Requirements & Literature Survey
W2 → System Design & Architecture
W3–4 → Backend & Database Development
W5–6 → AI Skill Matching
W7 → Frontend & Integration
W8 → Testing & Evaluation **W9** → Deployment & Documentation